

Chapter 6: Time and Frequency Characterization of Signals and Systems

Department of Electrical & Electronic Engineering

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Outline

- Magnitude and phase of Fourier transform
- First and second order filters
- Non-ideal low-pass filter

Magnitude & Phase

- Magnitude of spectrum, $|X(j\omega)|$ or $|X(e^{j\omega})|$, determines the energy of frequency component
- Phase of spectrum affects the shape of time-domain signal

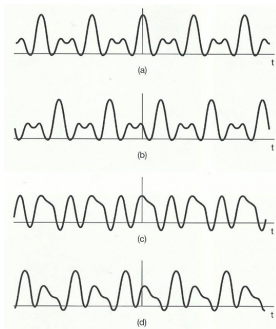


Figure: $x(t) = 1 + \frac{1}{2}\cos(2\pi t + \phi_1) + \cos(4\pi t + \phi_2) + \frac{2}{3}\cos(6\pi t + \phi_3)$

- Case by case, the magnitude and phase information of Fourier transform may carry information of different importance.

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Example: Magnitude & Phase of Image

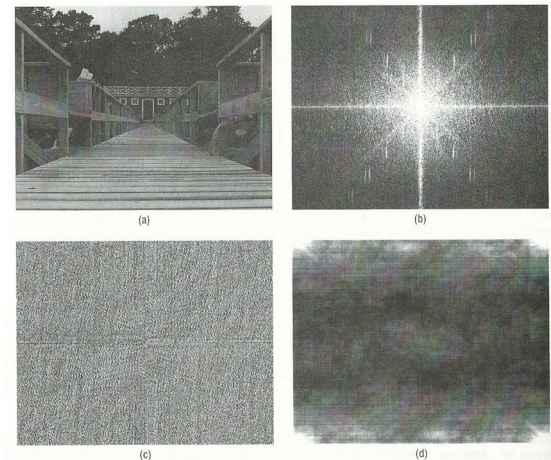


Figure: (a) Original image; (b) Magnitude; (c) Phase; (d) Set phase to zero

Example: Magnitude & Phase of Image

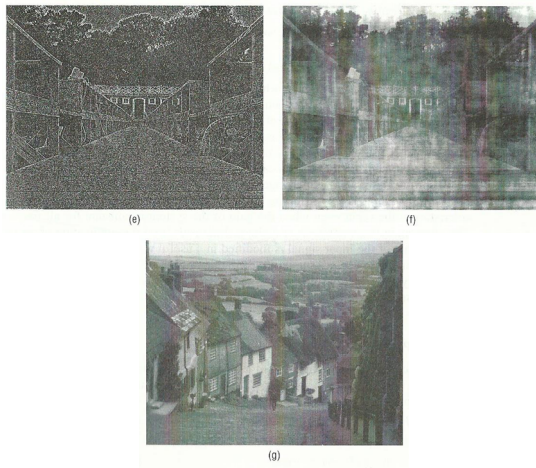


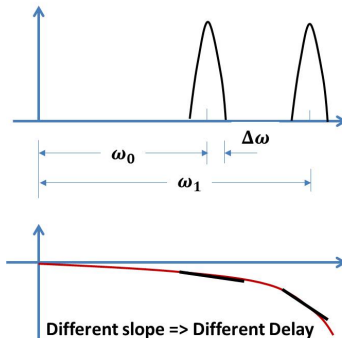
Figure: (e) Set magnitude to 1; (f) Original phase + (g)'s magnitude

LTI Systems: Linear & Nonlinear Phase

- Linear phase response leads to delay

$$X(j\omega)H(j\omega) = X(j\omega)e^{-j\omega t_0} \longleftrightarrow x(t - t_0)$$

- Non-linear phase response leads to distortion: $e^{-j\omega^2 t_0}$
- Narrow-band signal: phase response can be approximated as linear



Group Delay

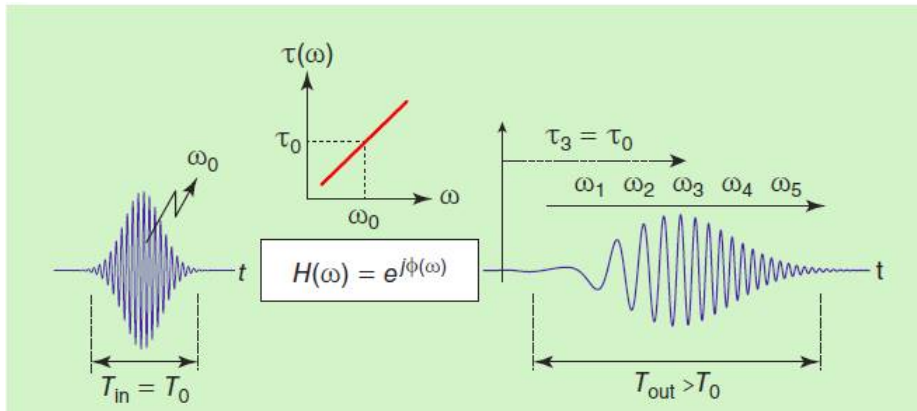
Suppose two narrow-band signals X_0 and X_1 are delivered into a system $H(j\omega) = e^{-j\omega^2 t_0}$:

$$-j\omega^2 t_0 \approx -2j\omega_0 t_0 \omega + j\omega_0^2 t_0$$

$$\begin{aligned} X_0(j\omega)H(j\omega) &= X_0(j\omega)e^{-j\omega^2 t_0} \\ &\approx X_0(j\omega)e^{-2j\omega_0 t_0 \omega} e^{j\omega_0^2 t_0} \\ &\leftrightarrow e^{j\omega_0^2 t_0} x_0(t - 2\omega_0 t_0) \end{aligned}$$

$$X_1(j\omega)H(j\omega) \leftrightarrow e^{j\omega_1^2 t_0} x_1(t - 2\omega_1 t_0)$$

Group delay: different frequency components have different delay



First-Order Continuous-Time Systems

- Differential equation:

$$\tau \frac{dy(t)}{dt} + y(t) = x(t) \quad \tau > 0 \quad (\text{why positive?})$$

- Frequency response:

$$H(j\omega) = \frac{1}{j\omega\tau + 1}$$

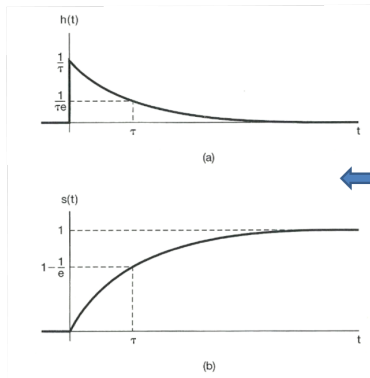
- Impulse response:

$$h(t) = \frac{1}{\tau} e^{-t/\tau} u(t)$$

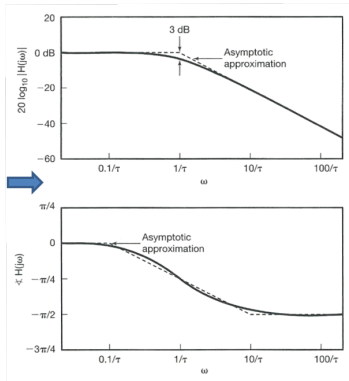
- Step response:

$$s(t) = [1 - e^{-t/\tau}] u(t)$$

First-Order Continuous-Time Systems: Plots

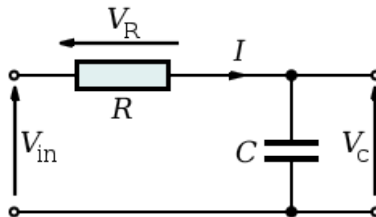


Trade-off



$$\text{Magnitude: } 20 \log_{10} |H(j\omega)| \approx \begin{cases} 0 & \text{if } \omega < 1/\tau \\ -20 \log_{10}(\omega) - 20 \log_{10}(\tau) & \text{if } \omega > 1/\tau. \end{cases}$$

Example: RC Circuit



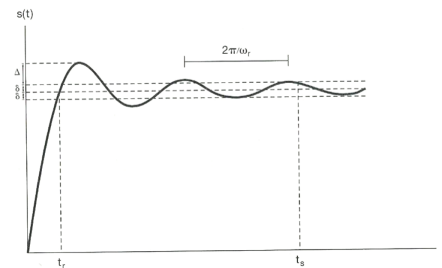
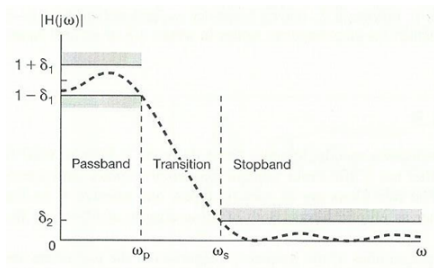
$$RC \frac{dV_c(t)}{dt} + V_c(t) = V_{in}(t)$$

$$\frac{V_c(j\omega)}{V_{in}(j\omega)} = \frac{1}{RCj\omega + 1} \text{ and } \frac{V_R(j\omega)}{V_{in}(j\omega)} = 1 - \frac{V_c(j\omega)}{V_{in}(j\omega)} = \frac{RCj\omega}{RCj\omega + 1}$$

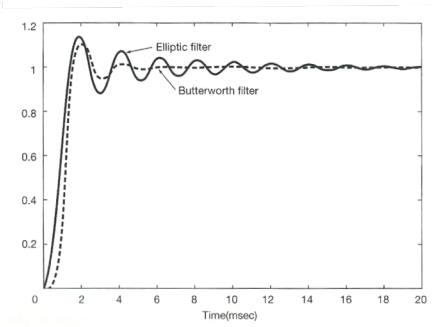
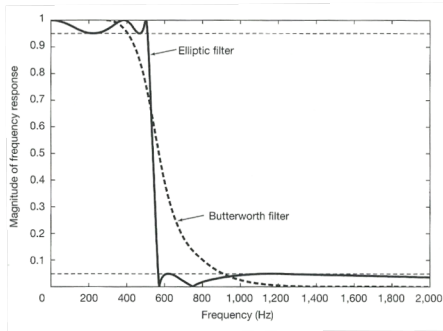
Implementation of adjustable first-order filter

Non-Ideal Low-pass Filter

- Ideal low-pass filter: non-causal, sharp edge, implementation issue
- Non-ideal filter: passband (with tolerant ripple) $\rightarrow$ transition band $\rightarrow$ stopband
- Step response: rise time, overshoot, ringing frequency, setting time



Example: Elliptic Filter v.s. Butterworth Filter

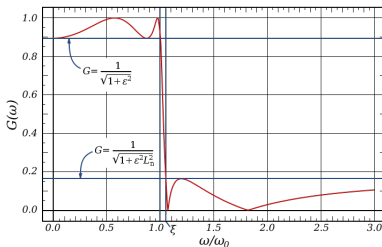


- See textbook, Example 6.3

Elliptic Filter

$$G_n(j\omega) = \frac{1}{\sqrt{1 + \epsilon^2 R_n^2(\xi, \omega/\omega_0)}}$$

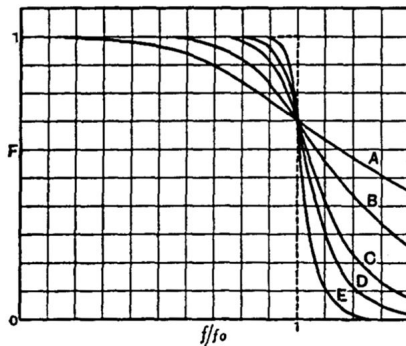
- where R_n is the n th-order elliptic rational function, ω_0 is the cut-off frequency, ϵ is the ripple factor, ξ is the selectivity factor
- "No other filter of equal order can have a faster transition in gain between the passband and the stopband, for the given values of ripple"



Butterworth Filter

$$G_n(j\omega) = \sqrt{\frac{1}{1 + \omega^{2n}}}$$

- "have as flat a frequency response as possible in the passband"
- Maximally flat magnitude filter



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